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Data flywheel: What it is and how it works

15–19 minutes

How Do AI Data Flywheels Work?

AI data flywheels work by creating a loop where AI models continuously improve by learning from the latest institutional knowledge and user feedback. As the system interacts with the environment, it collects feedback and new data, which are then used to refine and enhance the backbone models powering the AI workflows. This process involves curating and improving the quality of the data collected, using that to refine the underlying models, ensuring that the AI model's accuracy and performance are consistently enhanced while optimizing the total cost of ownership (TCO) for running these AI workflows.

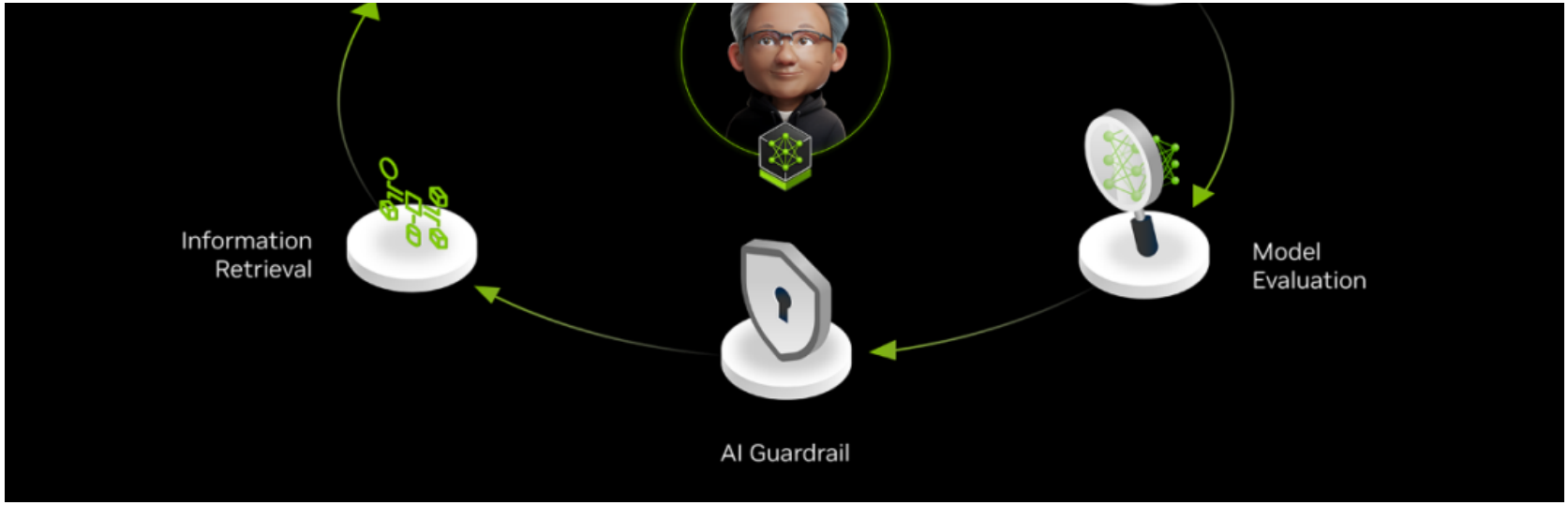

Figure 1: Data Flywheel: A continuous cycle of data processing, model customization, evaluation, guardrails, and deployment that uses enterprise data to improve AI systems

Additionally, AI guardrails are in place to maintain the integrity and reliability of the agent interaction, ensuring that the outputs are accurate, compliant, and secure. This continuous cycle of feedback and enhancement makes the AI systems increasingly effective over time.

This workflow process involves six key steps:

1. Data Processing**:** An AI data flywheel starts with enterprise data, which takes many forms—including text documents, images, videos, tables, and graphs. For an AI data flywheel, data processing is required to extract and refine raw data. The raw data is further filtered to remove noise, personally identifiable information (PII), and toxic or harmful data to curate high-quality data.

2. Model Customization**:** Using large language model (LLM) techniques like domain adaptive pretraining (DAPT), LoRA, and supervised fine-tuning (SFT), you can add domain-specific knowledge and task-specific skills to the model to build a deeper understanding of the company's unique vocabulary and context.

3. Model Evaluation**:** Then, you can evaluate the model's performance to verify the answers (outputs) align to application requirements. These first three steps are done iteratively to ensure that the model's quality is improved and the results are satisfactory for the intended use case.

4. **AI Guardrails Implementation:** Adding AI guardrails to your customized model ensures that enterprises' specific privacy, security, and safety requirements are met when the application is interacting with users and the environment.

5. **Custom Model Deployment:** There is often a need to continuously retrieve information at runtime from an expanding set of data sources. Implementing an effective retrieval-augmented generation (RAG) system ensures that the AI has access to the most relevant context, enabling it to serve the intended use case more accurately and efficiently.

6. **Enterprise Data Refinement:** With the wealth of domain expertise and enterprise data, the AI system continuously interacts with its environment, generating inference logs and capturing human and AI feedback. As a result, the institutional data is continuously updated over time with new data collected. This feeds back into the data processing step as the process is repeated to iteratively optimize the AI workflow for both cost efficiency and accuracy.

The core value lies in consistently capturing user feedback and system behavior. With an AI data flywheel in place, you can deliver smarter, more contextual responses, improve accuracy, and even distill smaller, more efficient models—all based on real-world usage patterns, ultimately optimizing TCO.

What Is the Purpose of a Data Flywheel Strategy for Scaling AI?

Real-world AI agent systems may have hundreds to thousands of AI agents simultaneously working collectively to automate processes. An AI data flywheel is imperative to streamline agent operations

(e.g., reviewing new data), especially as business requirements change. This guarantees smoother AI agent orchestration, as a specialized team of AI agents can provide resource-optimized plans and execute on those plans with minimal human input.

Agentic AI scalability depends on an automated cycle of data curation, model training, deployment, and institutional knowledge collection and review to improve the intelligent agents' performance.

In addition, AI applications involve a number of human collaborators with specific responsibilities:

Role	Responsibility
Data engineers	Must curate structured and unstructured data to generate high-quality data for training AI models
AI software developers	Must take the curated datasets to train the AI model further for specialized purposes
IT and MLOps teams	Must deploy the model in a safe environment while considering usage and access requirements
Human-in-the-loop and AI systems	Must review the institutional knowledge generated and make consistent adjustments to the database, as it is continuously fed back into the data engine

An automated data flywheel, powered by an end-to-end platform, can streamline the entire process of continuous improvement. It reduces the need for manual effort, minimizes tool integration overhead, and lowers the burden of human reviews. This setup makes it easier to maintain and scale AI systems —especially when building complex multi-agent systems.

Why Are Data Flywheels Important for Agentic and Generative AI Workflows?

When adopting AI agent and generative AI applications, a data flywheel is needed to drive the continuous improvement and adaptability of your application. As business requirements change or grow in complexity, performance and cost often become a differentiating factor for success.

For example, take AT&T, one of the world's largest telecommunications companies. As demand for personalized, always-on customer support continues to grow, AT&T is scaling AI-powered agents across its operations to deliver faster, more accurate customer service. Facing challenges like model drift, rising computational demands, and the need for real-time data access, AT&T leveraged NVIDIA NIM™ and NeMo™ microservices to build a data flywheel-driven AI platform that continuously improves performance while optimizing cost, speed, and compliance.

With an effective AI data flywheel, organizations can:

- Develop cost-effective applications tailored to business or customer needs.
- Personalize and refine offerings to improve user experiences.
- Meet tangible goals, such as increasing sales conversions or implementing automations to improve productivity.

To maintain a competitive edge, organizations can gather and process new interaction data, refine their AI models, and progressively enhance their AI applications' performance. From LLMs to vision language models (VLMs), a variety of data can be integrated.

Additionally, development teams can deploy smaller, more efficient models in production to significantly reduce the TCO for AI workflows. In some cases, building data flywheels has resulted in over 98% savings in inference costs—all without compromising accuracy.

This approach can significantly reduce the time and resources required to develop and deploy agentic and generative AI solutions.

When Should You Automate and Accelerate Your Data Flywheel?

Accelerating your data flywheel for AI is necessary to address dependencies associated with agentic AI technology.

For example, without a centralized system for feedback and logging, it's difficult to track and analyze system performance, which can slow down the data flywheel. Evaluation datasets that don't accurately reflect real-world scenarios can lead to models that perform poorly.

As knowledge bases are updated, the relevance of system feedback can decline, making it harder for the flywheel to continuously improve. Human intervention, while beneficial, is resource-intensive and time-consuming. Addressing this is crucial for accelerating the data flywheel and maintaining its effectiveness.

As such, acceleration becomes necessary when many interactions are happening on a system level that impact performance. For example, in generative AI applications, accuracy and alignment to human preferences are important. In agentic AI applications, streamlined plans and execution of those plans by AI knowledge workers are required.

Operational Requirements	Recommendation
Facilitating resource-intensive tasks, such as training data review	With centralized user data collection and automatic insight generation, user data classification and triaging streamlines human-in-the-loop review.

Operational Requirements	Recommendation
Enhancing agentic AI and generative AI applications by refining models	A data flywheel can be powered with a Helm chart deployment or via API calls for specific parts of your workflow.
Running secure deployments and protecting enterprise data	Running end-to-end workflows on a GPU-accelerated cloud or private data center provides higher security, privacy, control, and integration flexibility.

How Does a Data Flywheel Prevent Model Drift and Outdated Responses?

When deploying an AI application, ensuring it provides up-to-date responses is a key design consideration.

As more and more data from the latest AI interaction is captured by your data flywheel setup, models can be continuously refined to prevent model drift and outdated responses. This can be done by creating a continuous feedback loop that captures real-world usage data, evaluates model performance, and triggers retraining or fine-tuning as needed. As users interact with the system, the flywheel collects high-signal data—such as incorrect predictions, low-confidence outputs, or evolving user behavior—and uses it to improve future model iterations. This ongoing refinement helps the model stay aligned with current user needs and context, reducing the risk of stale or inaccurate outputs over time.

At the same time, a well-designed data flywheel can lower the TCO by automating key parts of the model lifecycle, such as evaluation, data curation, and fine-tuning. Rather than retraining large models from scratch, the system can fine-tune smaller, optimized models using only the most relevant data,

preserving accuracy against the latest curated ground truth while significantly reducing compute and infrastructure costs. By combining smart data collection with efficient model refinement, the flywheel ensures long-term performance, reliability, and cost efficiency, all of which are essential for scalable AI in production.

How Can You Get Started With Data Flywheels?

Building the next generation of agentic AI and generative AI applications using a data flywheel involves rapid iteration and use of institutional data.

NVIDIA NeMo™ is an end-to-end platform for building data flywheels, enabling enterprises to continuously optimize their AI agents with the latest information.

NeMo helps enterprise AI developers easily curate data at scale, customize LLMs with popular fine-tuning techniques, consistently evaluate models on industry and custom benchmarks, and guardrail them for appropriate and grounded outputs.

An end-to-end platform powered by modular AIOps microservices exposed as simple-to-use API calls makes it convenient for development teams to automate and speed up model training and experimentation with their proprietary data to build data flywheels.

The NeMo platform includes:

- NeMo Curator**:** To efficiently curate high-quality datasets for training LLMs, thereby enhancing model performance and accelerating the deployment of AI solutions.
- NeMo Customizer**:** A high-performance, scalable microservice that simplifies the fine-tuning and alignment of LLMs with popular parameter-efficient fine-tuning techniques including LoRA and DPO.
- NeMo Evaluator**:** An enterprise-grade microservice that provides industry-standard benchmarking of

generative AI models, synthetic data generation, and end-to-end RAG pipelines.

- NeMo Guardrails**:** A microservice for developers to implement robust safety and security measures in LLM-based applications, ensuring that these applications remain reliable and aligned with organizational policies and guidelines.
- NeMo Retriever**:** A collection of microservices that drive the AI data flywheel by enabling scalable data ingestion and high-accuracy, privacy-preserving retrieval. With fast, context-aware responses from large data collections, developers are able to connect AI applications to diverse data sources, build AI query engines, and continuously refine AI models with real-time insights.

  Alternatively, the AI Data Flywheel NVIDIA Blueprint, powered by the NVIDIA NIM and NeMo microservices, provides a reference architecture for a quick start to building data flywheels. This blueprint enables teams to continuously distill LLMs into smaller, cheaper, and faster models without compromising accuracy, using real-world production traffic from the AI agent interactions. It automates the execution of structured experiments that explore the space of available models, surfacing promising efficient candidates for production promotion or deeper manual evaluation.